

Implementing Simple Linear Regression Using Python

In this article, let's look at the concept of linear regression with the help of an example and learn about its implementation in Python, and more.

Machine learning (ML) is a scientific study of algorithms and statistical models that computer systems use, in order to perform a specific task effectively without using explicit instructions.

Supervised learning is a type of ML in which a machine learns from labelled data and predicts the output from those learnings. Linear regression is a popular model that comes under supervised learning. Basically, regression algorithms are used for predicting continuous variables like stock prices, weather forecasts, etc.

What is linear regression?

Regression refers to the search for relationships among variables. We can relate it to the sales of ice-cream, which are determined by the weather of the city. With an increase in humidity, ice-cream sales also rise.

In simple terms, linear regression is used for finding the relationship between

two variables — the input variable (x) and the output variable (y). A single output variable can be calculated from a linear combination of multiple input variables. Input variables are also known as independent variables or predictors, while output variables are also known as dependent variables or responses.

Let us look at the simplest algorithm used for prediction.

Learning model: Ordinary least squares (OLS)

This common method can have multiple inputs. The ordinary least squares procedure seeks to minimise the sum of the squared residuals. Each time, we calculate the distance from every data point to the regression line, square it, and take the sum of all the squared errors together, ending up with considering the parameters of the least sum of all squared errors. The following example elaborates this by predicting the price of wine, given its age.

Consider the output variable (y) as the price and the input variable (x) as the age of the wine. Do remember that there could be more than one input variable. To help understand this better, we will run through an example with the values shown in Table 1.

Age (in years)	Price (Rs)
0	75
1.2	100
2.1	220
3.4	300
4.1	440
5.6	500
7	700

Table 1: The age (x) and price (y) of wine

Visualisation

The graphical representation of the data can be seen in Figure 2.

```
# Visualising the Training set
plt.scatter(X, y, color = 'red')
plt.title('Age vs Price of Wine
(Training set)')
plt.xlabel('Price of Wine')
plt.ylabel('Age of Wine')
plt.show()
```

We can easily see that there is a relationship between these two variables. Technically, this suggests that the variables are correlated. The correlation coefficient 'R' determines how strong the correlation is.

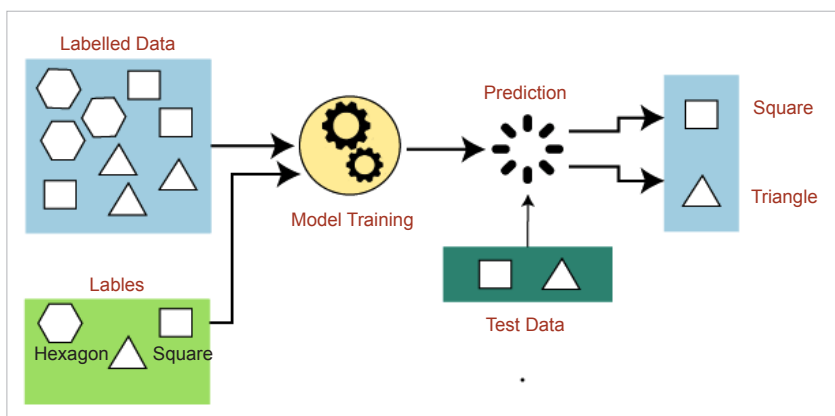


Figure 1: Supervised learning (Credits: <https://www.javatpoint.com/supervised-machine-learning>)